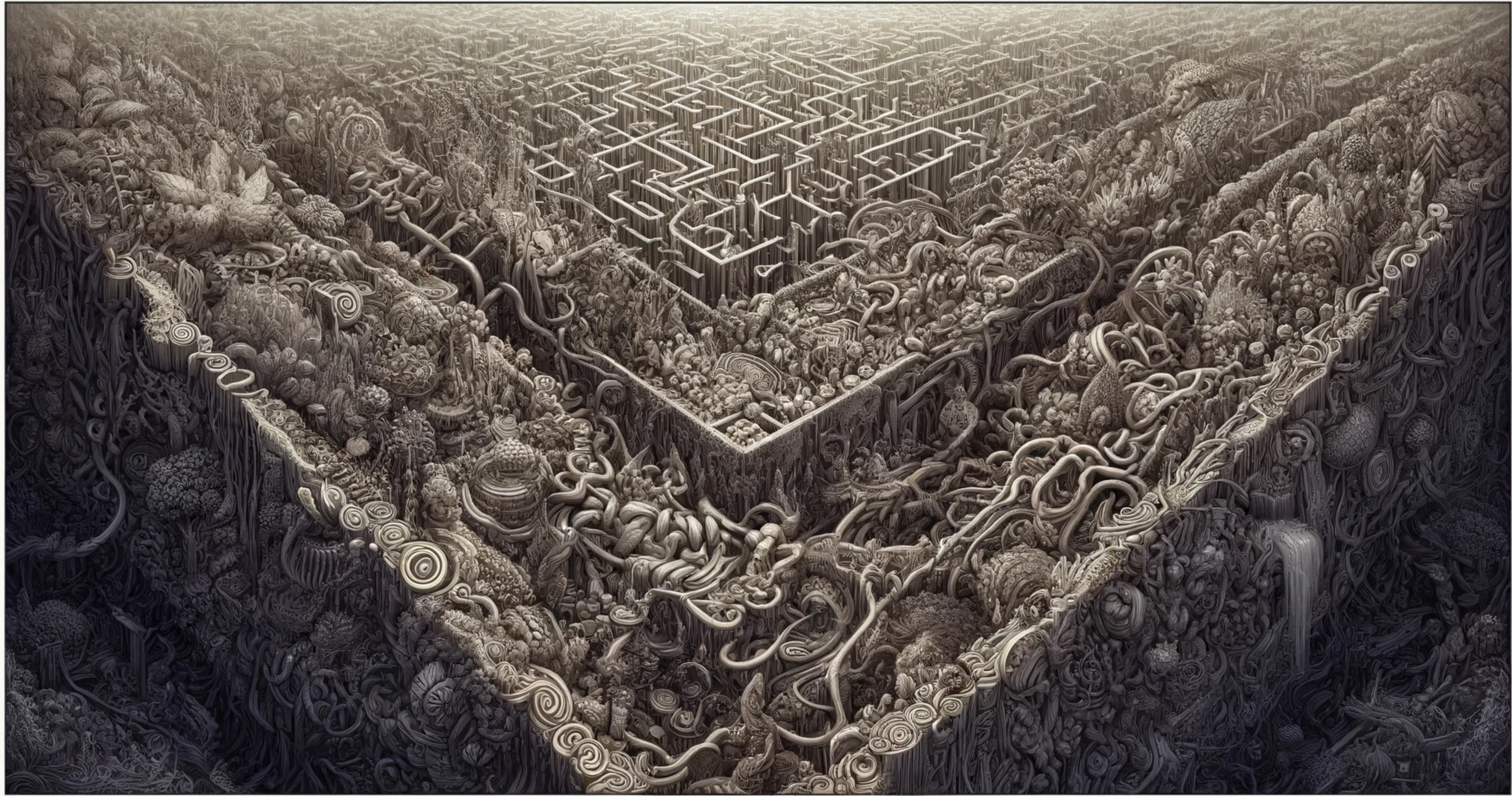


Data Analysis & Visualisation

Raoul Grouls, 8 September 2026



Lounen mnc(hd)' • 0111ce | Rtsurel
Geineahioeemethod |  To/tonle level

Some maps are less useful...



...than others

Model of the Earth, Leonardo Da Vinci

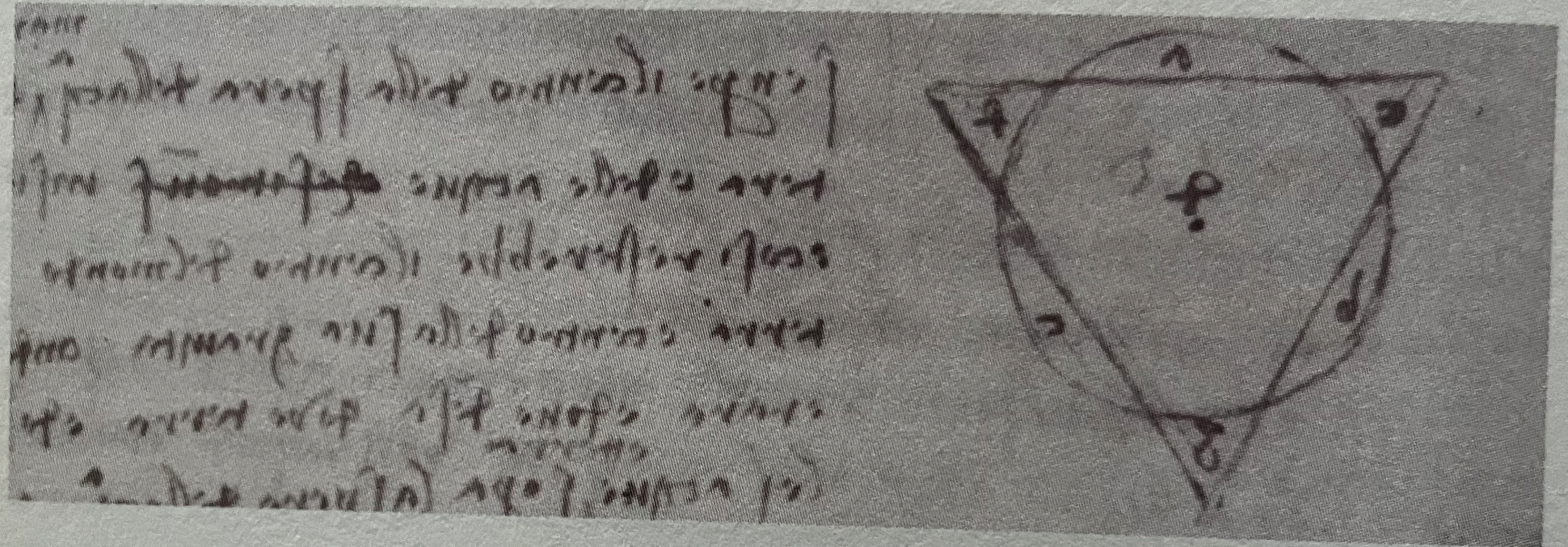


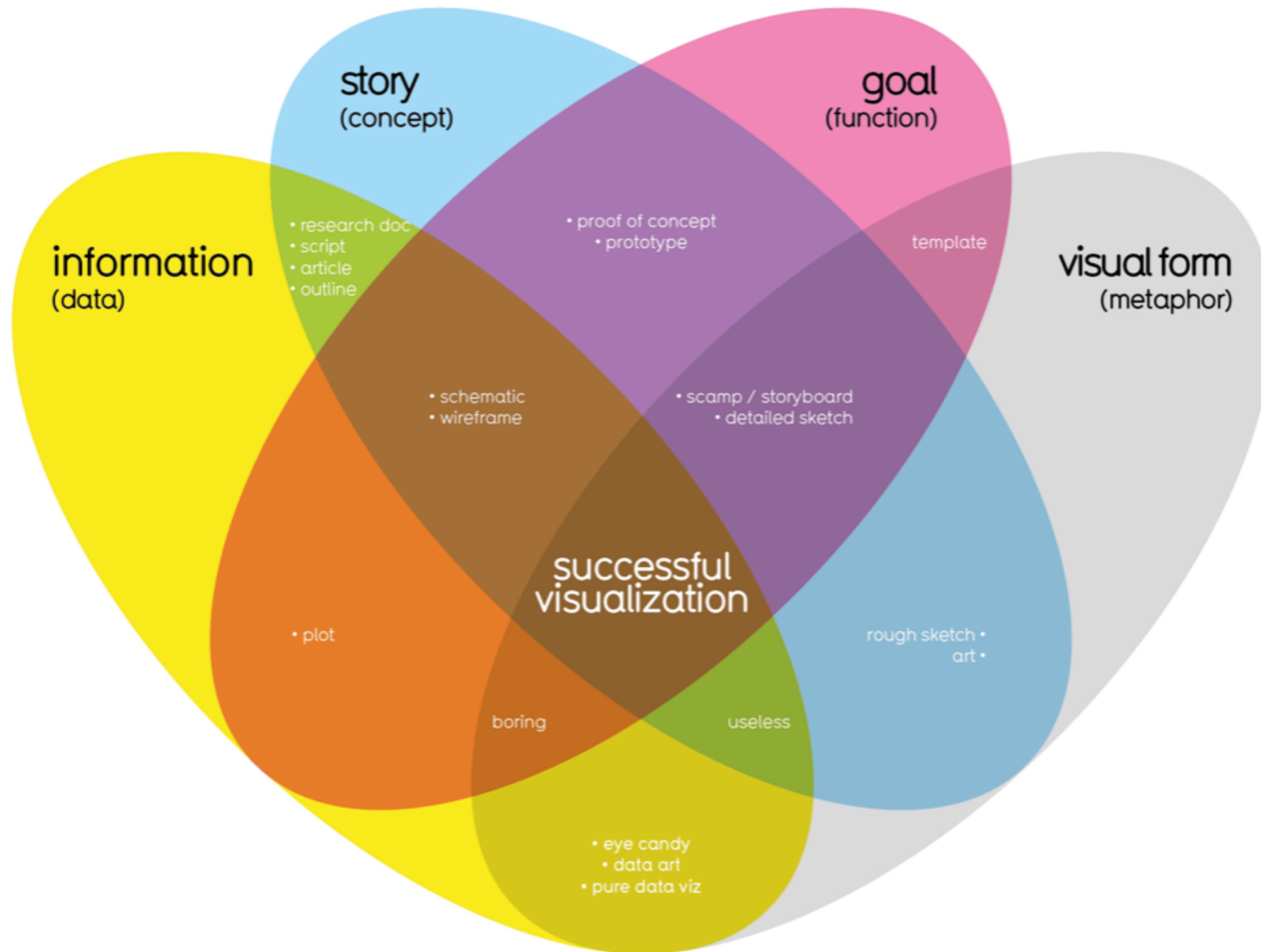
FIG. 2-7. Geometric model of the Earth.
Codex Leicester, folio 35v (detail).

Models don't need to be “correct” to be useful or insightful.
Just homomorphic enough

rollover for more detail

What Makes a Good Visualization?

explicit (implicit)



Gestalt principles

GESTALT PRINCIPLES

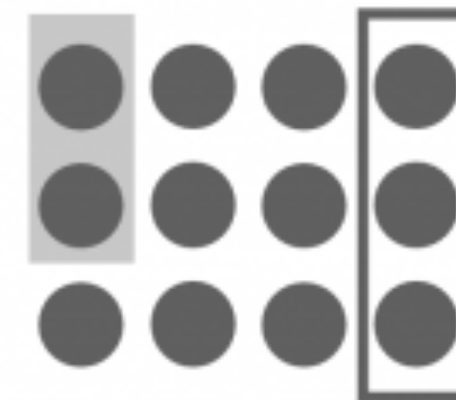
Proximity



Similarity



Enclosure



Connection



Continuity



Closure



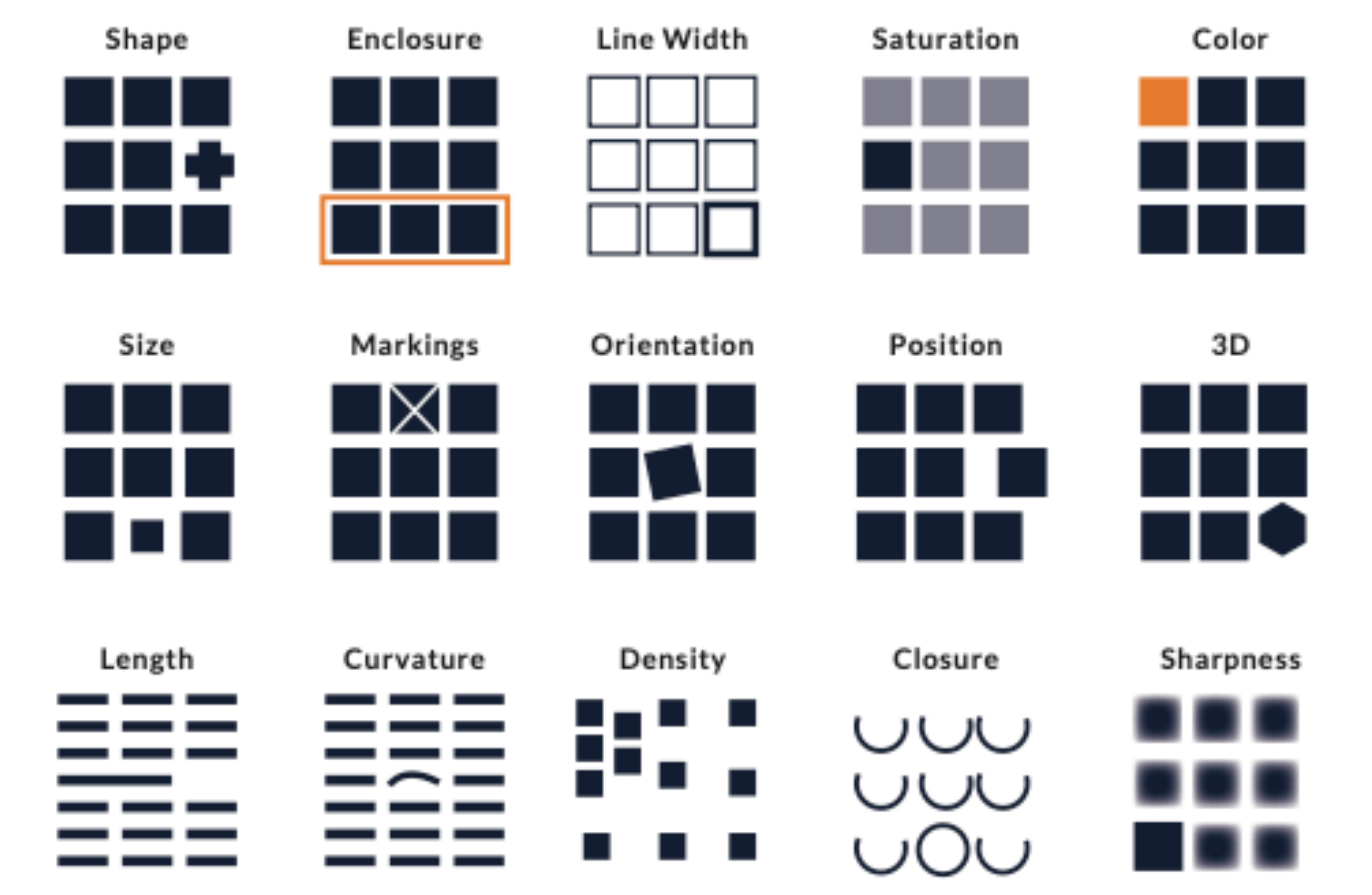
Figure-Ground



Common Fate

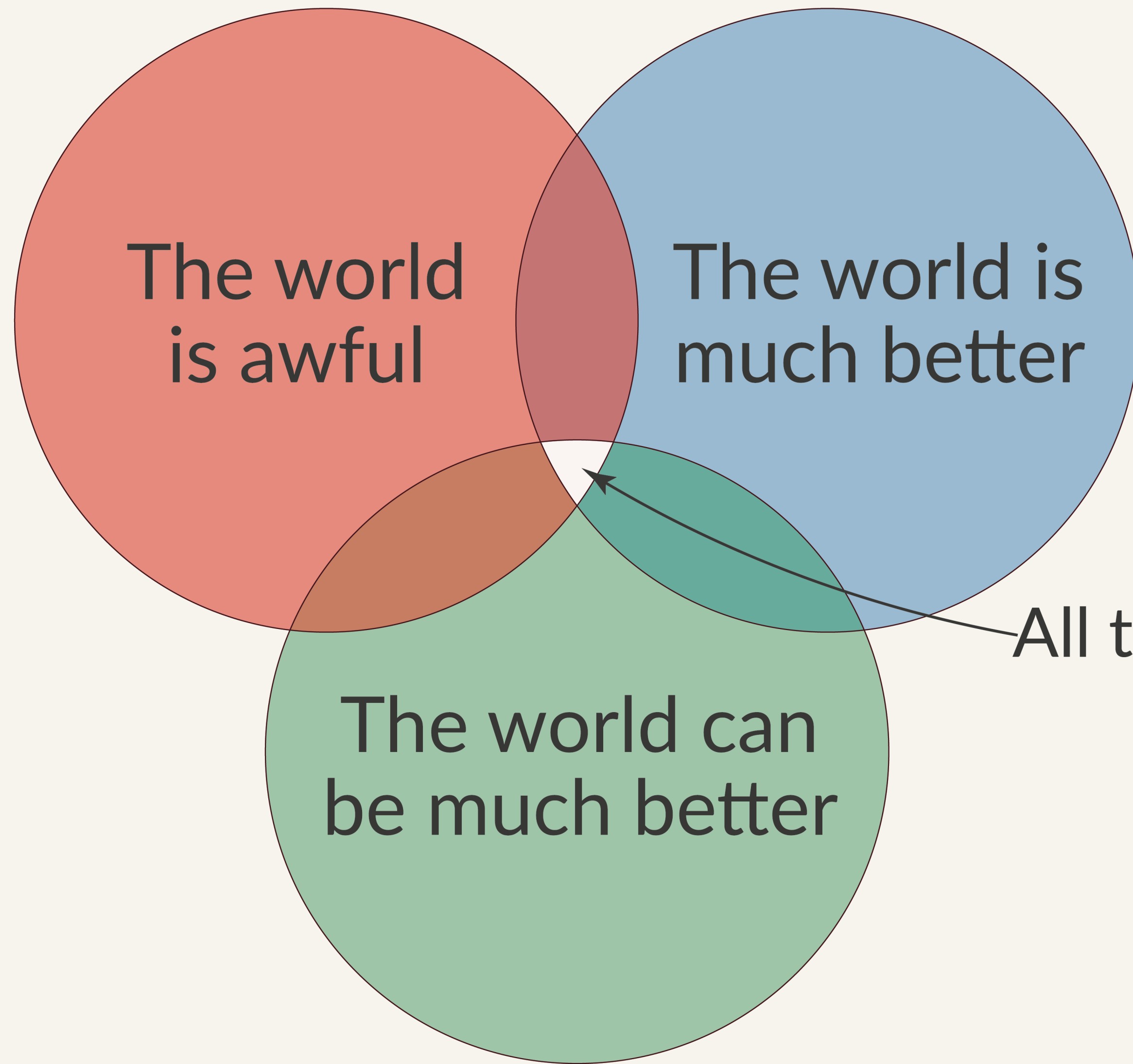


Preattentive processing



Five guidelines

0. Avoid spurious correlations
1. Show the data
2. Reduce the clutter
3. Integrate graphics and text
4. Avoid the spaghetti chart
5. Start with grey



All three statements are true at the same time

The world is awful

Our World
in Data

Globally 4.3% of all children die before they are 15 years old.

This means every day 16,000 children die.

*Each rectangle represents
the death of 11 children.
That's the number of children
who die every minute.*



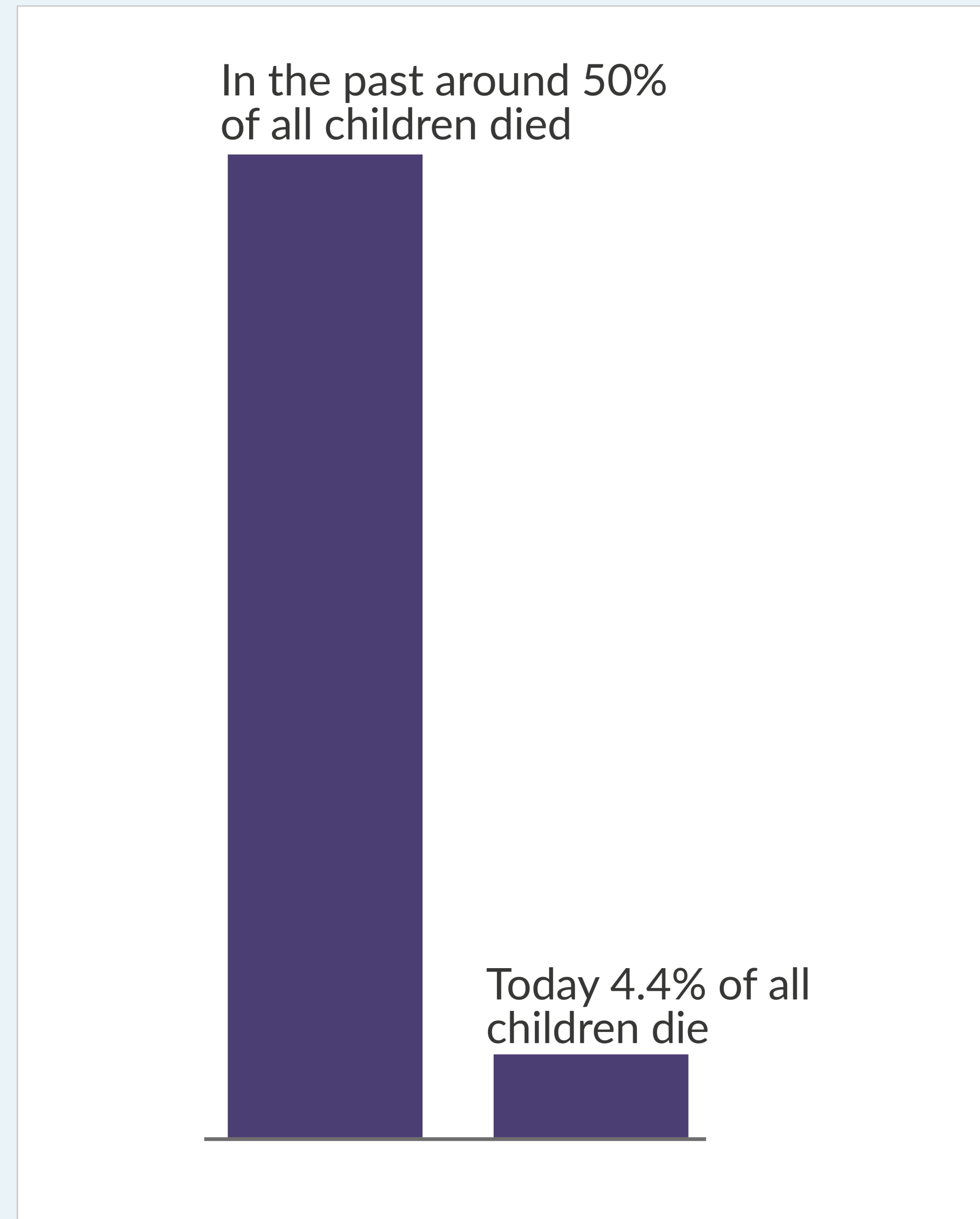
Data: 2020 data from UN IGME.

[OurWorldinData.org](https://ourworldindata.org) – Research and data to make progress against the world's largest problems.

Licensed under CC-BY by the author Max Roser

The world is much better

Our World
in Data



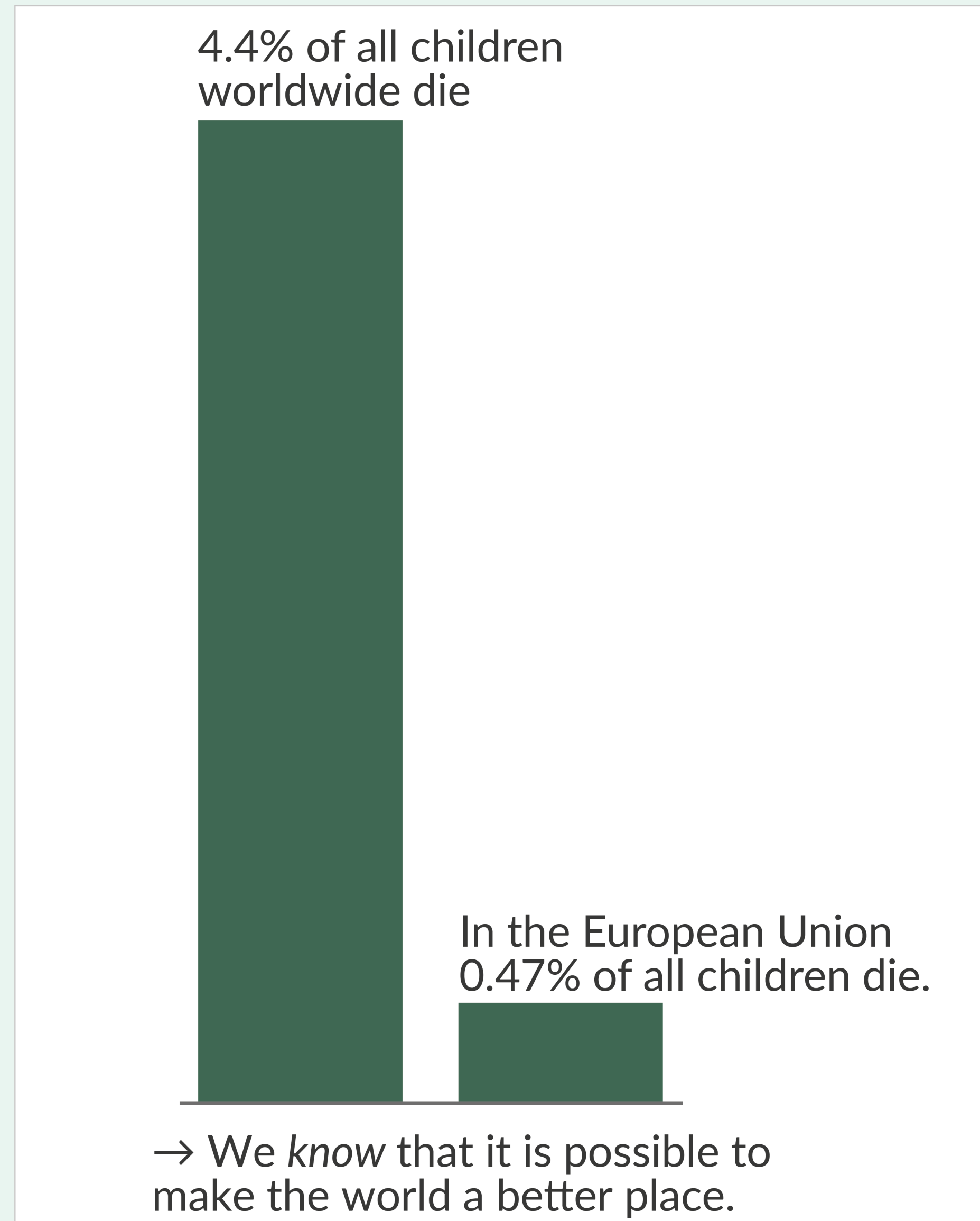
Data: 2020 data from UN IGME and historical data from Volk and Atkinson (2013).

OurWorldinData.org – Research and data to make progress against the world's largest problems.

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The world can be much better

Our World
in Data



Data: 2020 data from UN IGME.

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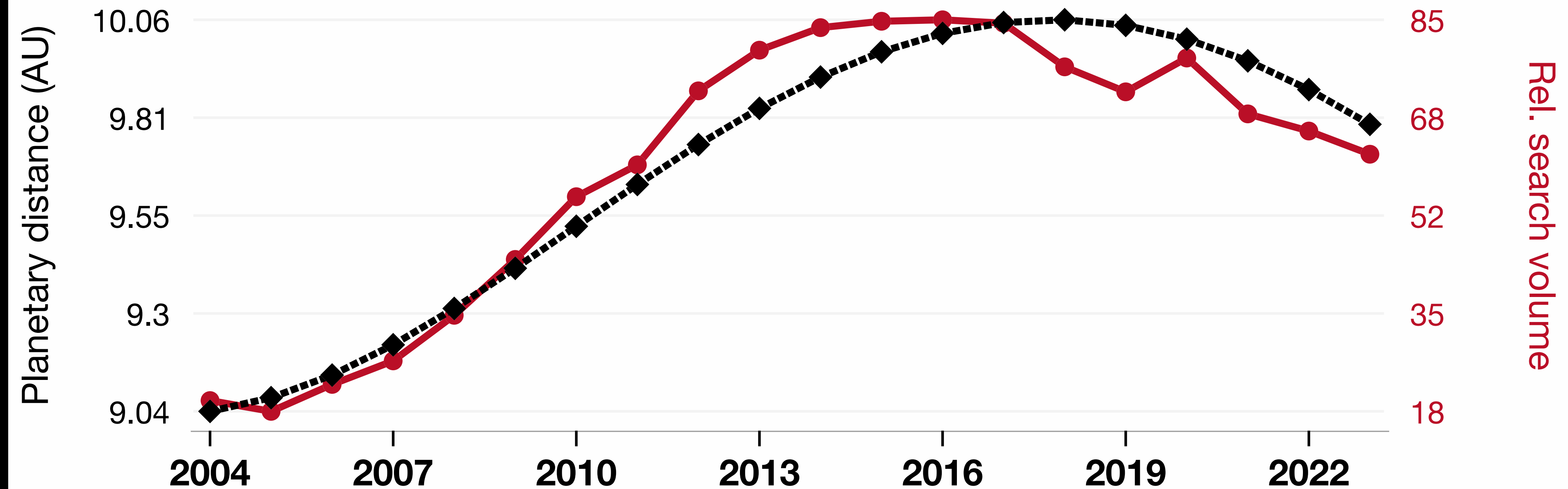
OurWorldinData.org – Research and data to make progress against the world's largest problems.

Spurious correlations

The distance between Saturn and the Sun

correlates with

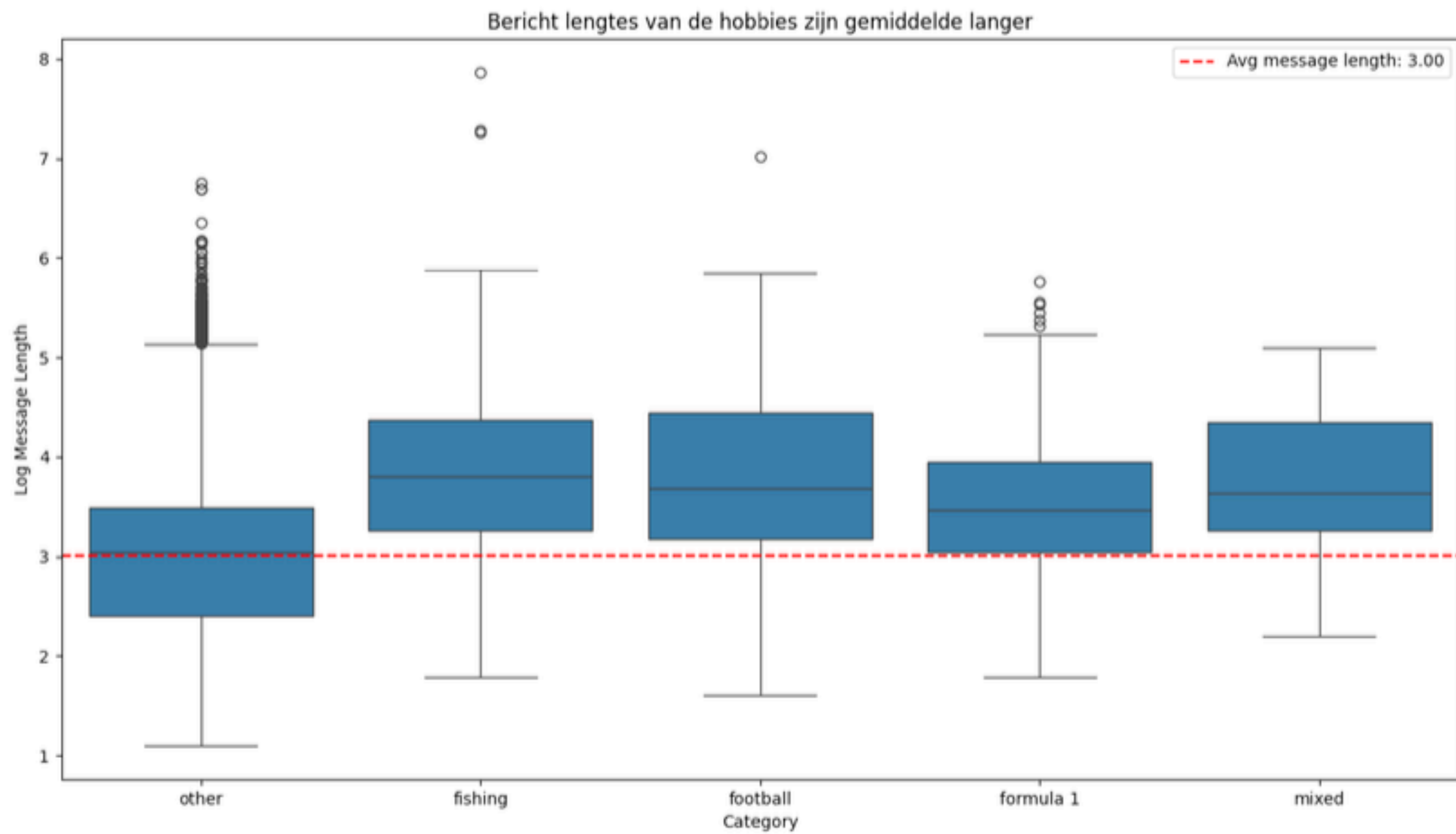
Google searches for 'how to make baby'



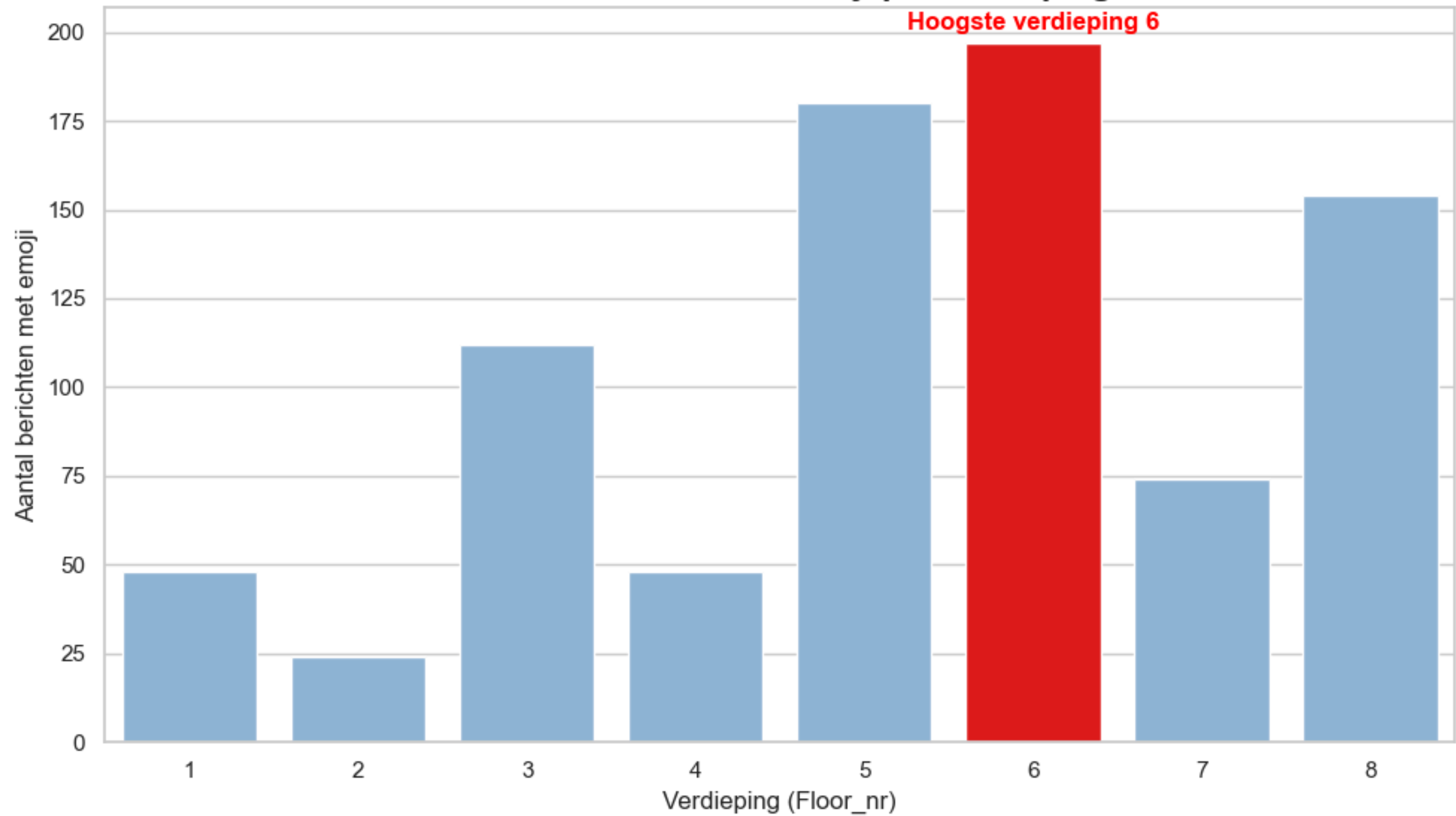
◆ The average distance between Saturn and the Sun as measured on the first day of each month · Source: Cacculated using Astropy

● Relative volume of Google searches for 'how to make baby' (Worldwide, with quotes) · Source: Google Trends

2004-2023, $r=0.967$, $r^2=0.935$, $p<0.01$ · tylervigen.com/spurious/correlation/1522



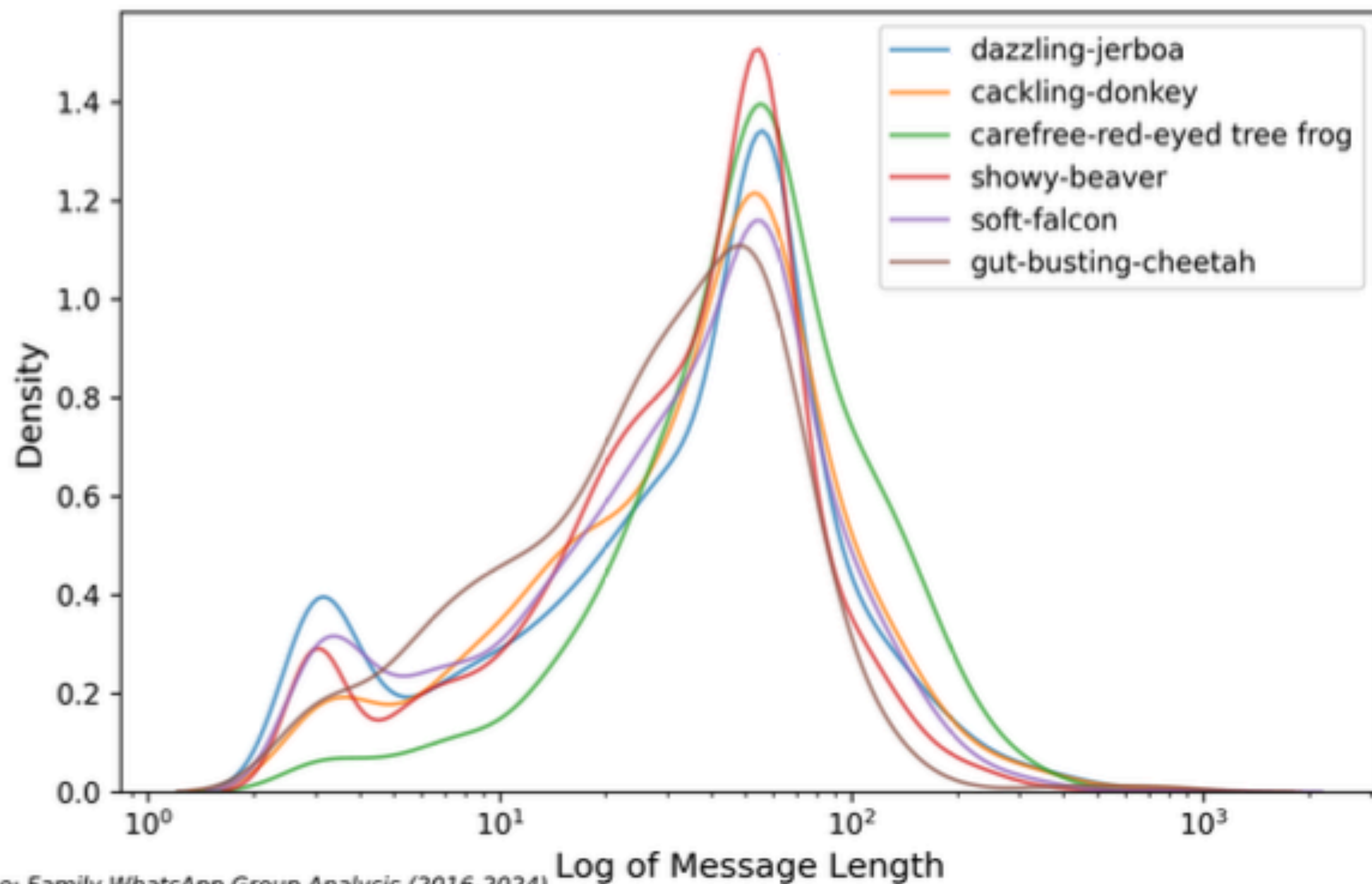
Aantal berichten met emoji per verdieping



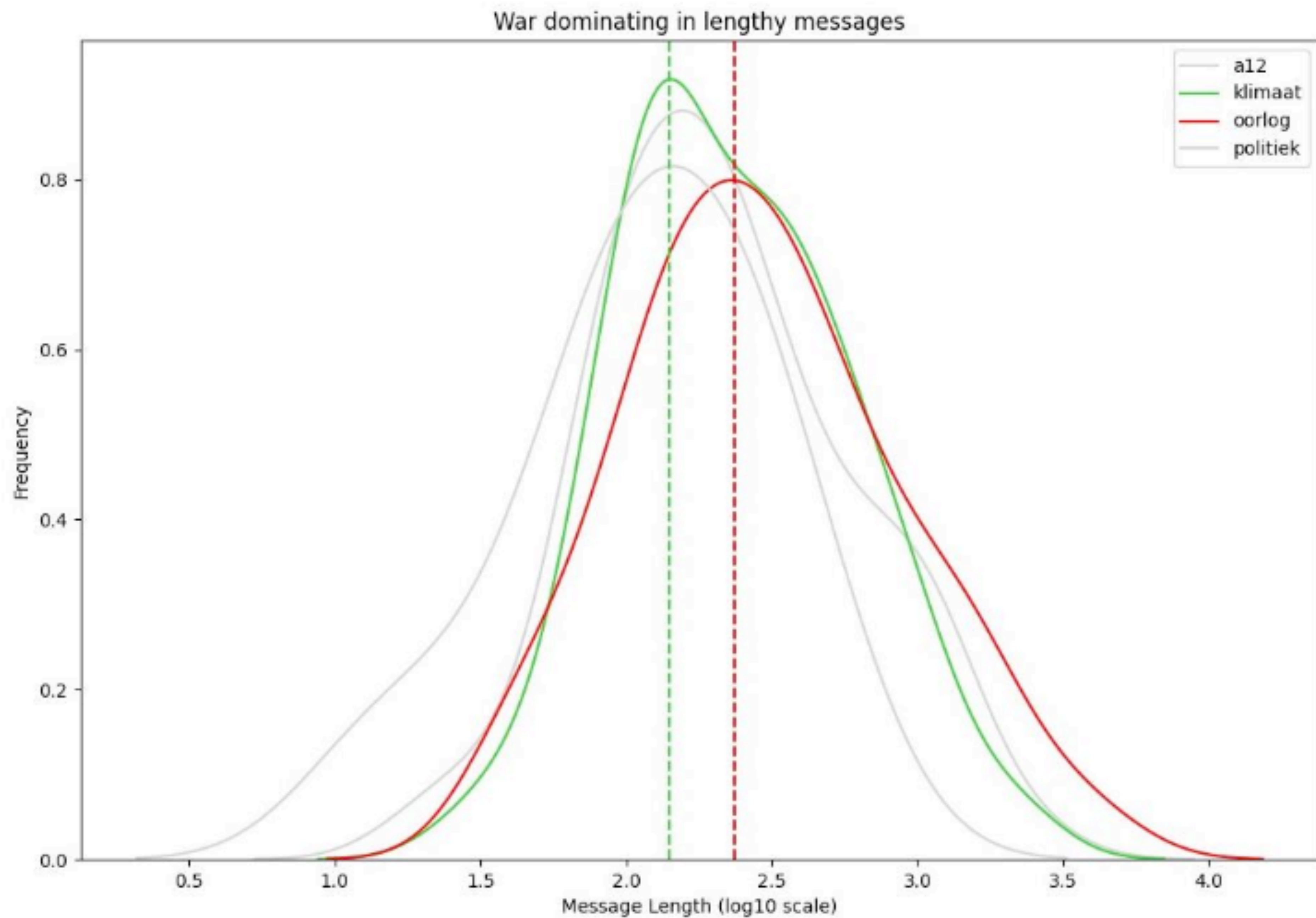
Convincing patterns

Authors' Distinct Messaging Styles Unveiled

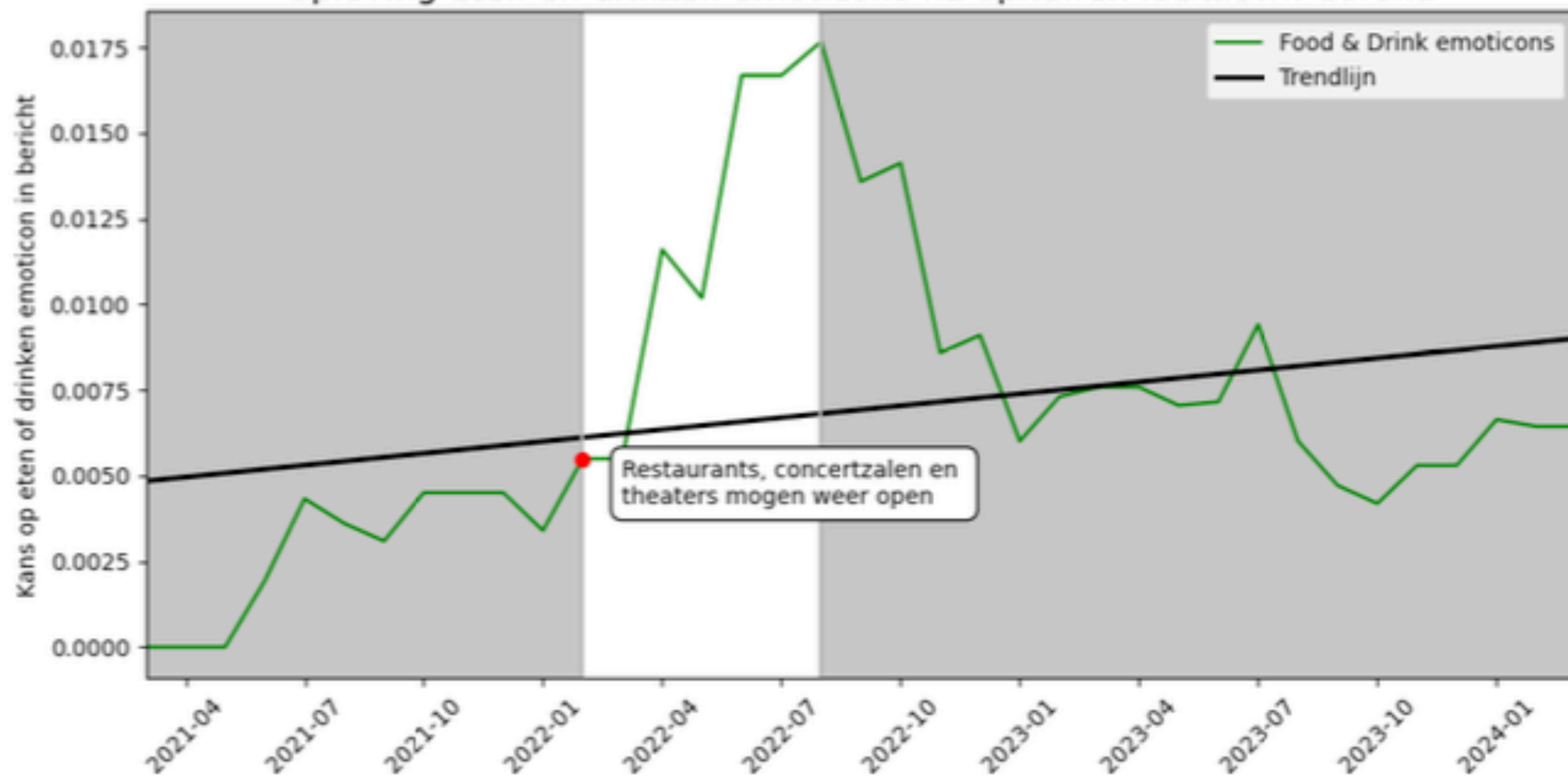
Each author's messaging habits are distinct, showcased through the peaks and patterns in message length



Source: Family WhatsApp Group Analysis (2016-2024)



Opleving eten en drinken emoticons na opheffen lockdown Corona



Import Ouddorp spreekt echt een andere taal

